

Experimental Study of Phenakite Solubility in Aluminosilicate Melts: Implication for the Genesis of Be Deposits

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Abstract—The solubility of phenakite (Be_2SiO_4) in granite melts was experimentally studied at temperatures of 1000 and 1100°C and pressures of 1 and 4 kbar in dry conditions and in the presence of 10 wt % H_2O . The starting materials were granite glasses with apaitic coefficient of 1–2.5 and natural phenakite. It was found that the solubility of phenakite increases with increasing apaitic index $(\text{Na} + \text{K})/\text{Al}$ of the melt, and the solubility of BeO in hydrous melts is higher than in dry ones. The solubility of phenakite also increases with pressure. The obtained experimental data were generalized with the previous data in the form of an equation describing the solubility of BeO in alkaline-granite melts coexisting with crystalline Be phases depending on the apaitic index, temperature, and pressure. The results of the experiments and their generalizations support the model of Be concentration in alkaline hydrous melts—the products of differentiation of granite magmas.

Keywords: phenakite, beryllium, aluminosilicate melt, experiment

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INTRODUCTION

Beryllium is a typical rare element ascribed to trace elements. The Be clark varies from 3 ppm in the upper crust to 60–100 ppb in primitive mantle (Grew, 2002). In spite of the low abundance, Be forms about 30 own minerals. The Be content increases with increase of silica of magmatic rocks. Its maximum clarkes were established in granites, where its average content in some massifs reaches 130–160 ppm, although usually varies within 3–28 ppm (London and Evensen, 2002). The average Be content in the alkaline rocks corresponds to that of granites.

In magmatic processes, Be as incompatible element is accumulated mainly in residual pegmatite melts, and is redistributed in hydrothermal fluids in the presence of fluorine. Correspondingly, two main genetic types of Be deposits are distinguished: pegmatite and hydrothermal (Kupriyanova and Shpanov, 2011). In the ores of economic deposits, the average Be content varies from 700 to 7200 ppm (0.2–2 wt % BeO). Among hydrothermal deposits, the largest reserves were found in the deposits related to the ber-

trandite—fluorite—phenakite (Ermakovskoe, Russia), bertrandite—argillite (Spor Mountain, USA), beryl—micaceous (Malyshevskoe, Russia), and rare metal—fluorite (Voznesenskoe, Russia) metasomatites (*Geneticheskie tipy ...*, 1975; Kupriyanova and Shanov, 2011; Lykhin and Yarmolyuk, 2015; Barton and Young, 2002). Majority of hydrothermal beryllium deposits associate with relatively small granite and syenite bodies (Kupriyanova and Shpanov, 2011). For instance, one of the world's largest Ermakovskoe F—Be deposit in Western Transbaikalia is genetically related to small stock of alkaline granites, although F.G. Reyf (2008) suggests that this massif is an inlier of the larger unexposed intrusion.

Since all beryllium deposits are genetically related to granitoid magmatism, the accumulation of this element is caused, first of all, by processes related to the generation and differentiation of parental aluminosilicate melts. However, data on the beryllium solubility in melts are very scarce. Only few experimental studies are available on the beryllium behavior in magmatic and hydrothermal processes (Beus and Dikov, 1967;

Evensen et al., 1999; London et al., 1988; Wood, 1992). The most complete experimental results on the Be solubility in the water-bearing aluminosilicate melts are listed in the work (Evensen et al., 1999), where the solubility of beryl ($\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$) and phenakite (Be_2SiO_4) in melts was studied within a temperature range of 650–850°C at 2 kbar. The authors revealed that the Be solubility increases with increasing temperature and depends on the index $\text{ASI} = \text{Al}_2\text{O}_3/(\text{Na}_2\text{O} + \text{K}_2\text{O} + \text{CaO})$ of the melt. However, experiments (Evensen et al., 1999) were carried out for melts with agpaitic index $K_{\text{agp}} = (\text{Na} + \text{K})/\text{Al} < 1$. The influence of alkalinity on the phenakite solubility (Be_2SiO_4) in the aluminosilicate melts was studied by Damdinov et al. (2021) at 1000°C and 1 kbar. Experiments at higher pressures and temperatures were not carried out.

Experimental works raises question about the possible accumulation of high Be concentrations for the formation of large deposits in relation with small intrusions. It is known that Be in the hydrothermal processes is mainly accumulated in fluoride and fluorine–carbonate complexes (Wood, 1992). Factors responsible for the behavior of this element in magmatic process have not been established reliably yet. Since practically all hydrothermal Be deposits associate with rocks of elevated alkalinity, the solubility of this element, in addition to temperature and pressure, is determined by the alkalinity and acidity of melts. The need for the study of the influence of indicated factors on the Be behavior served as the basis for the widening of conditions of experimental studies of phenakite (Be_2SiO_4) solubility in the aluminosilicate melts of different alkalinity at different temperatures and pressures, results of which are reported in this paper.

EXPERIMENTAL METHOD

To study the dependence of phenakite solubility in the aluminosilicate melts of different alkalinity on P – T parameters, experimental series was carried out at 1000 and 1100°C and pressures of 1 and 4 kbar in dry conditions and in the presence of 10 wt % H_2O (Suk et al., 2024a; Suk et al., 2024b). The experiments were carried out on a high gas pressure vessel with internal heating. The starting material was granite from the Orlovka deposit, Transbaikalia (hole 42), of the following composition (wt %): SiO_2 —72.10; TiO_2 —0.01; Al_2O_3 —16.14; Fe_2O_3 —0.68; MnO —0.09; CaO —0.30; MgO —0.01; Na_2O —5.17; K_2O —4.28; P_2O_5 —0.02; F —0.32; H_2O —0.18. Mixtures with different agpaitic index (1–2.5) were prepared from granite powder by addition of corresponding amounts of K_2CO_3 and Na_2CO_3 . They melted in platinum crucibles at 1250°C in a KO-14 furnace for 12 h and quenched in glasses of different alkalinity. Grains of natural phenakite 0.5–1 mm in size from the Ermakovskoe F–Be deposits

were used for experiments on dissolution. The experiments were conducted in welded ampoules 3 mm across, which were loaded with an charge of granite glass + phenakite and added with water, if required. The experiments lasted for five days. Experimental conditions are listed in Table 1.

The compositions of run samples were determined on a Tescan Vega II XMU scanning electron microscope (Czech Republic) equipped with energy dispersive (INCAx-sight) and crystal diffraction (INCA wave) spectrometers (England, Oxford) at the Institute of Experimental Mineralogy, Russian Academy of Sciences, using an INCA Energy 450 software for the qualitative and quantitative analysis. Operation conditions of the energy dispersive spectrometer are as follows: accelerating voltage of 20 kV, beam current of Co-absorbed electrons of 0.3 nA, and counting time of 70 s. The measurement accuracy of the energy dispersive spectrometer is 0.2 wt %.

The contents of Be and trace elements were determined by laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) on a high-resolution Element XR (Thermo Fisher Scientific) mass spectrometer coupled with a UP-213 laser device with a wavelength of 213 nm (New Wave Research) at the Geospektr Center for Collective Use of the Dobretsov Geological Institute of the Siberian Branch of the Russian Academy of Sciences (Ulan Ude). A NIST-610 glass standard was used as external standard, and a NIST-612 glass was used as the reference sample. The external and reference samples were measured each five measurements.

Data were processed using a Glitter software (Griffin et al., 2008) with introduction of SiO_2 determined by electron probe microanalysis as an internal standard. The relative measurement errors were less than 10% for Be, <15% for Na, <7% for Al, <10% for K, and <8% for Ca.

EXPERIMENTAL RESULTS

Experimental products. The products of all experiments were homogenous glass with phenakite crystals (Fig. 1). In experiments with melts with agpaitic index >1.5 , the drop-like and vermicular segregations and small crystals supposedly of BeO were observed around phenakite crystals (Fig. 2). Figure 2b demonstrates a complete decomposition of phenakite crystals, which indicates an intense removal of SiO_2 from phenakite according to reaction $\text{Be}_2\text{SiO}_4 \rightarrow 2\text{BeO} + \text{SiO}_2$ (melt).

Glass composition. The composition of glass (Tables 2 and 3) was determined mainly near crystals. However, the BeO content weakly changed over the entire sample, which indicates the attainment of equilibrium during experiments. The agpaitic index of the melts in the presence of water slightly decreases compared to the initial value, which is explained by the

Table 1. Conditions of experiments on study of phenakite solubility in model granite melts

Run number	Charge	T , °C	P , kbar	H_2O , μL	K_{agp} , ³⁾ init.
7286	70 mg granite ¹⁾ + 14 mg phenakite ²⁾	1000	1	7	1
7287	70 mg granite + 14 mg phenakite	1000	1	7	1.2
7288	70 mg granite + 14 mg phenakite	1000	1	7	1.5
7289	70 mg granite + 14 mg phenakite	1000	1	—	1
7290	70 mg granite + 14 mg phenakite	1000	1	—	1.2
7291	70 mg granite + 14 mg phenakite	1000	1	—	1.5
Be-1	60 mg granite + 11.5 mg phenakite	1100	1	—	1
Be-2	60 mg granite + 11.5 mg phenakite	1100	1	—	2
Be-3	60 mg granite + 12 mg phenakite	1100	1	—	2.5
Be-4	60 mg granite + 12 mg phenakite	1100	1	6	1
Be-5	60 mg granite + 12.5 mg phenakite	1100	1	6	2
Be-6	60 mg granite + 12 mg phenakite	1100	1	6	2.5
Be-7	60 mg granite + 12 mg phenakite	1100	4	—	1
Be-8	60 mg granite + 12 mg phenakite	1100	4	—	2
Be-9	60 mg granite + 12 mg phenakite	1100	4	—	2.5
Be-10	60 mg granite + 12 mg phenakite	1100	4	8	1
Be-11	60 mg granite + 12 mg phenakite	1100	4	8	2
Be-12	60 mg granite + 12 mg phenakite	1100	4	8	2.5
Be-13	60 mg granite + 13 mg phenakite	1000	4	—	0.75
Be-14	60 mg granite + 15 mg phenakite	1000	4	—	1.25
Be-15	60 mg granite + 20 mg phenakite	1000	4	—	1.5
Be-16	60 mg granite + 13 mg phenakite	1000	4	8	0.75
Be-17	60 mg granite + 15 mg phenakite	1000	4	8	1.25
Be-18	60 mg granite + 20 mg phenakite	1000	4	8	1.5
Be-19	60 mg granite + 15 mg phenakite	1100	1	—	1.5
Be-20	60 mg granite + 15 mg phenakite	1100	4	—	1.5

(μL)—microliter. ¹⁾ model glasses of granite composition; ²⁾ phenakite Be_2SiO_4 , fraction 0.5–1 mm; ³⁾ K_{agp} —initial agpaitic index— $(Na + K)/Al$.

partial redistribution of K and Na in a fluid. The BeO content in glasses varies depending on the experimental conditions and glass composition from 0.34–0.67 wt % in dry conditions at 1000°C to 7.0–7.32 wt % in hydrous melts at 1100°C.

The phenakite solubility expressed in the BeO concentration in glasses strongly depends on the melt composition, increasing with a growth of its agpaitic index $K_{agp} = (Na + K)/Al$ (Fig. 3). Thereby, the solubility of phenakite in the presence of water fluid is higher than in dry conditions (Fig. 3). It also increases with an increase of pressure (Figs. 3, 4). In particular, at $K_{agp}=1.5$ in a dry system, the BeO content in the glass increases from ~1.30 wt % at 1 kbar to ~2.32 wt % at 4 kbar at 1000°C and, respectively, from ~1.25 wt % to ~2.52 wt % at 1100°C. At constant pressure, the growth of temperature shows no significant influence on the phenakite solubility in dry melts with $K_{agp} < 1.5$ at 1 kbar and < 2 at 4 kbar. More alkaline

melts at $K_{agp} = 1.5$ –2 show an increase of BeO content in the melt (Fig. 4) with the growth of temperature, which is well expressed in the water-bearing experiments. With a temperature increase (at 4 kbar and $K_{agp} = 1.50$) from 1000 to 1100°C, the BeO concentration in the melt increases from ~3 to ~5.5 wt %.

Obtained data on the dependence of phenakite solubility on the agpaitic index (K_{agp}) in the studied compositional interval are well described by the second-order polynomial equation:

$T = 1000^\circ C$, $P = 1$ kbar (hydrous system)

$$C_{BeO} (wt \%) = -1.14649 + 3.37594X - 0.68922X^2, \quad n = 3, \quad (1)$$

where $X = K_{agp}$; n —number of points;

$T = 1000^\circ C$, $P = 1$ kbar (dry system)

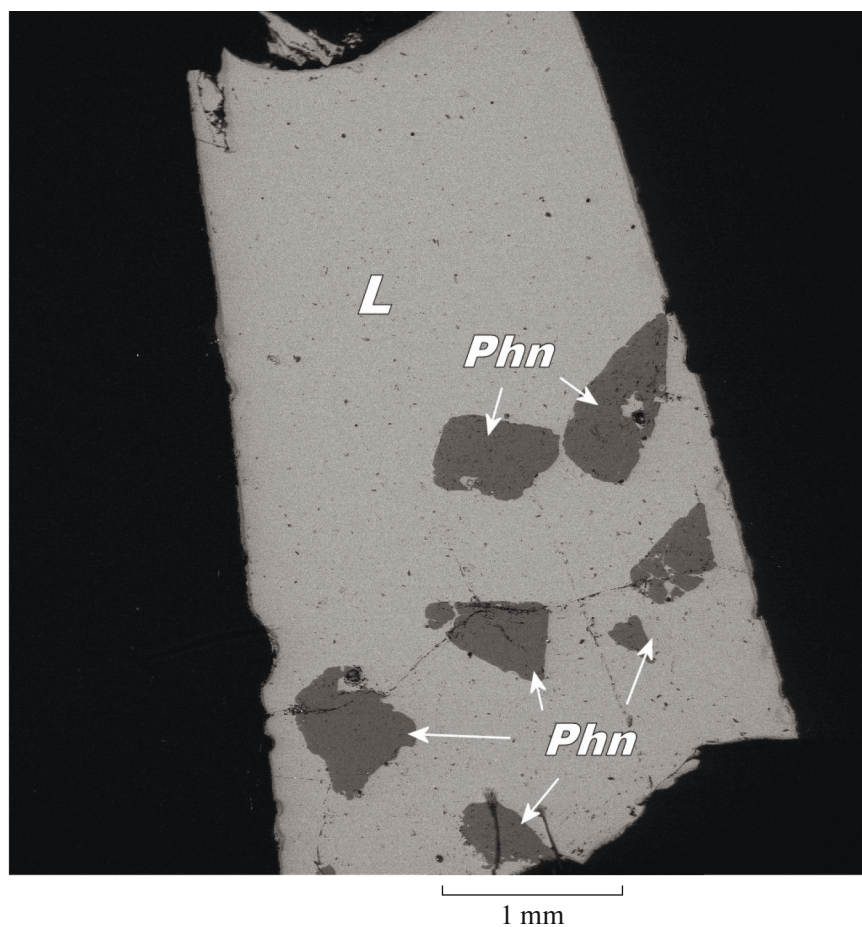


Fig. 1. Aluminosilicate glass with phenakite crystals (dark) after experiment at 1100°C and 1 kbar (sample Be-4). (*Phn*) phenakite, (*L*) melt.

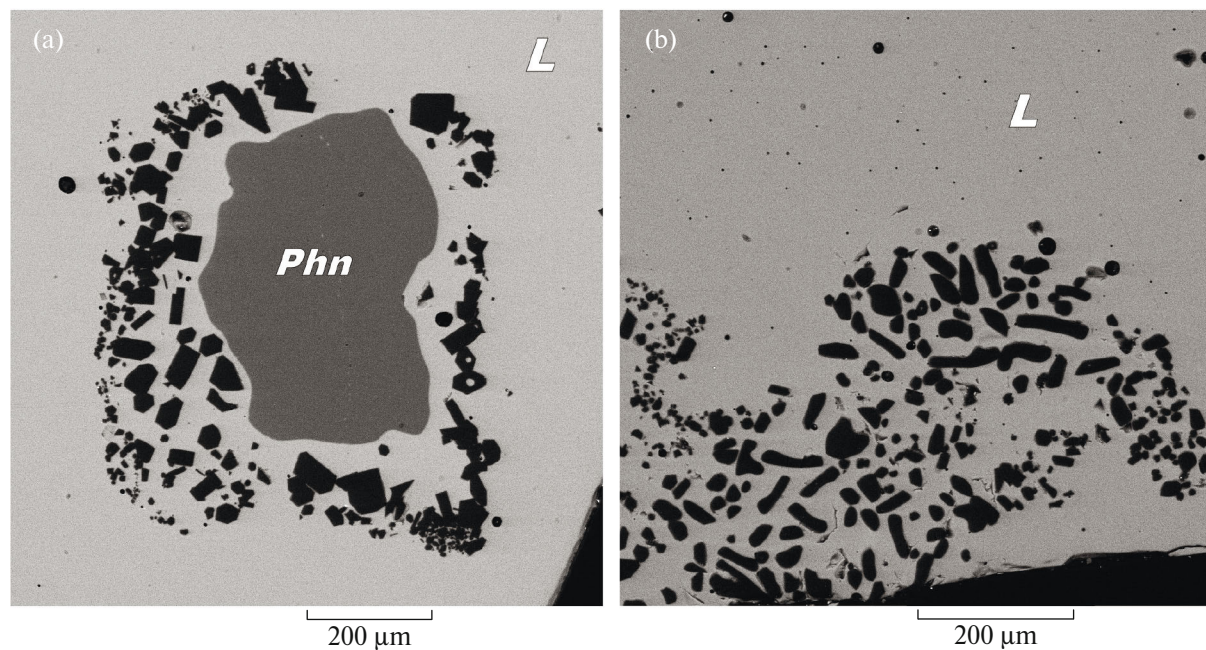


Fig. 2. Drop-like and vermicular segregations and small crystals supposedly of beryllium oxide formed around phenakite crystals (*Phn*) in water-bearing alkaline aluminosilicate melts. (a) sample Be-18 (1000°C, 4 kbar), (b) sample Be-11 (1100°C, 4 kbar).

Table 2. Compositions of experimental glasses obtained at 1000°C

Oxides	$T = 1000^{\circ}\text{C}$ and $P = 1$ kbar						$T = 1000^{\circ}\text{C}$, $P = 4$ kbar					
	In the presence of H_2O			Dry conditions			Dry conditions			In the presence of H_2O		
	7286	7287	7288	7289	7290	7291	Be-13	Be-14	Be-15	Be-16	Be-17	Be-18
SiO_2	68.70	67.74	67.62	71.91	70.64	69.81	73.96	72.68	70.02	72.64	70.85	70.44
Al_2O_3	14.01	13.11	12.31	15.31	14.88	13.23	15.70	12.56	10.36	13.34	11.64	10.35
Na_2O	5.23	5.40	6.16	5.96	6.91	7.72	4.13	5.65	7.04	3.21	3.86	4.28
K_2O	4.81	5.35	6.18	5.23	5.74	6.84	5.42	6.25	7.28	4.42	5.51	6.25
CaO	0.45	1.44	0.40	0.57	0.36	0.62	0.27	0.81	2.03	0.47	1.11	1.06
FeO	0.30	0.19	0.17	0.35	0.42	0.48	0.18	0.28	0.13	0.10	0.12	0.01
BeO	1.50	1.77	2.17	0.67	1.05	1.30	0.34	1.76	3.15	0.82	1.90	2.60
Total	95.00	95.00	95.00	100.00	100.00	100.00	100.00	100.00	100.00	95.00	95.00	95.00
K_{agp}	0.98	1.12	1.36	1.01	1.18	1.52	0.81	1.28	1.88	0.75	1.06	1.33

Table 3. Compositions of experimental glasses obtained at 1100°C

Oxides	$T = 1100^{\circ}\text{C}$, $P = 1$ kbar							$T = 1100^{\circ}\text{C}$, $P = 4$ kbar						
	Dry conditions				In the presence of H_2O			Dry conditions				In the presence of H_2O		
	Be-1	Be-2	Be-3	Be-19	Be-4	Be-5	Be-6	Be-7	Be-8	Be-9	Be-20	Be-10	Be-11	Be-12
SiO_2	75.20	67.85	61.49	70.53	73.29	62.64	57.33	76.56	66.31	60.50	70.94	74.94	62.93	57.88
Al_2O_3	13.12	10.25	10.77	10.96	11.21	10.74	11.11	12.57	10.68	10.60	11.25	9.93	10.82	11.18
Na_2O	4.36	8.30	11.62	8.35	3.62	6.92	9.67	4.00	7.76	11.27	7.42	3.09	5.95	9.27
K_2O	5.31	8.86	9.56	6.97	4.82	8.16	9.13	5.46	9.21	9.78	6.39	4.41	8.25	9.11
CaO	1.02	1.31	0.63	0.63	0.48	0.85	0.41	0.49	0.63	0.81	1.27	0.55	0.76	0.50
FeO	0.24	0.05	0.03	0.11	0.06	0.02	0.02	0.12	0.00	0.09	0.02	0.06	0.13	0.07
BeO	0.75	3.38	5.90	2.45	1.51	5.67	7.32	0.79	5.41	6.95	2.72	2.01	6.15	7.00
Total	100.00	100.00	100.00	100.00	95.00	95.00	95.00	100.00	100.00	100.00	100.00	95.00	95.00	95.00
K_{agp}	0.99	2.27	2.74	1.94	1.00	1.88	2.32	0.99	2.13	2.75	1.70	0.99	1.73	2.25

$$C_{\text{BeO}} (\text{wt } \%) = -5.09294 + 8.67647X - 2.94118X^2, \quad n = 3;$$

$T = 1000^{\circ}\text{C}$, $P = 4$ kbar (hydrous system)

$$C_{\text{BeO}} (\text{wt } \%) = -3.01457 + 6.26527X - 1.53669X^2, \quad n = 3;$$

1000°C , $P = 4$ kbar (dry system)

$$C_{\text{BeO}} (\text{wt } \%) = -2.78998 + 4.39757X - 0.65851X^2, \quad n = 3;$$

$T = 1100^{\circ}\text{C}$, $P = 1$ kbar (hydrous system)

$$C_{\text{BeO}} (\text{wt } \%) = -4.60915 + 6.8595X - 0.74036X^2, \quad n = 3;$$

$$(2) \quad T = 1100^{\circ}\text{C}, P = 1 \text{ kbar (dry system)}$$

$$C_{\text{BeO}} (\text{wt } \%) = 2.17044 - 2.98373X + 1.57845X^2, \quad n = 4, \quad S_x = 0.12, \quad (6)$$

$$(3) \quad \text{where } S_x \text{—root-mean-square deviation;}$$

$$T = 1100^{\circ}\text{C}, P = 4 \text{ kbar (hydrous system)}$$

$$(4) \quad C_{\text{BeO}} (\text{wt } \%) = -8.91139 + 14.14312X - 3.14284X^2, \quad n = 3; \quad (7)$$

$$T = 1100^{\circ}\text{C}, P = 4 \text{ kbar (dry system)}$$

$$(5) \quad C_{\text{BeO}} (\text{wt } \%) = -2.9545 + 3.6596X - 0.00094X^2, \quad n = 4, \quad S_x = 0.47. \quad (8)$$

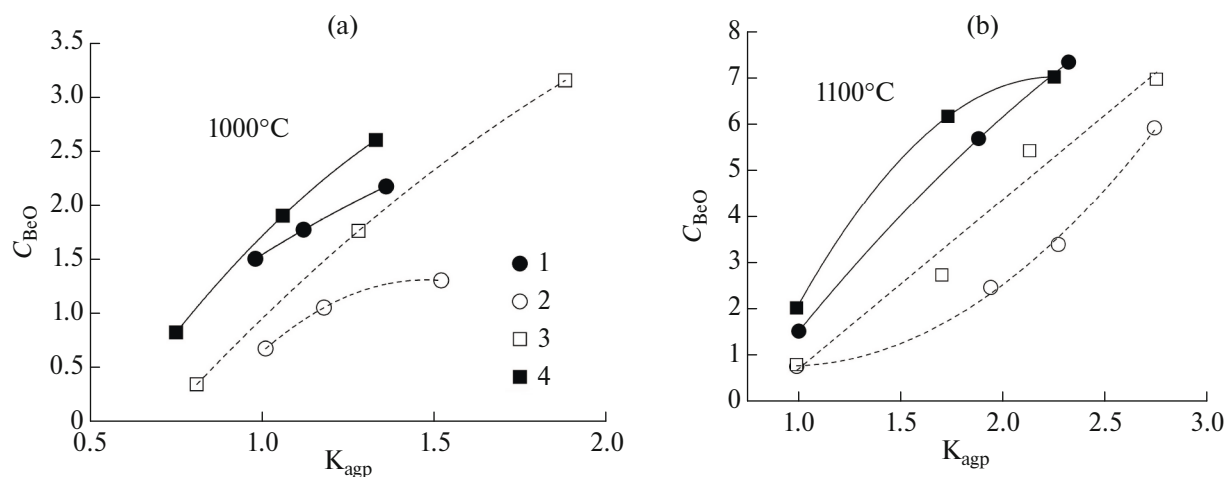


Fig. 3. Variations of the phenakite solubility versus agpaitic index in the aluminosilicate melt (wt %): (a) at 1000°C; (b) at 1100°C. (1) in water-bearing system, (2) in dry system at 1 kbar; (3) in dry system, (4) in water-bearing system at 4 kbar.

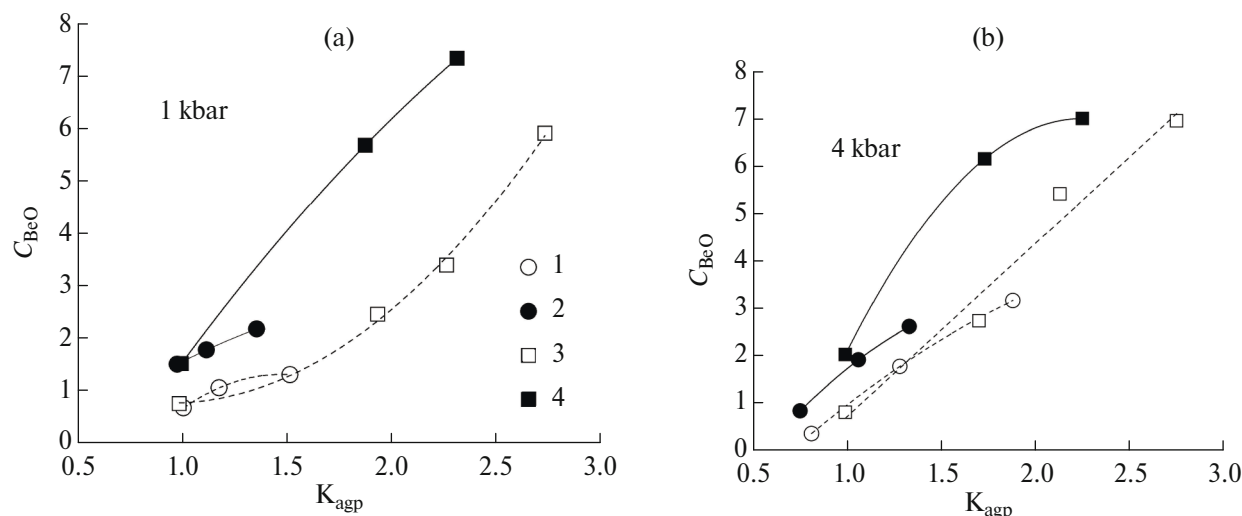


Fig. 4. Variations of the phenakite solubility in aluminosilicate melt (wt %) versus agpaitic index: (a) at 1 kbar; (b) at 4 kbar. (1) in dry system, (2) in water-bearing system at 1000°C; (3) in dry system, (4) in water-bearing system at 1100°C.

DISCUSSION

Beryllium Solubility in Silicate Melts

New data at temperatures of 1000 and 1100°C and pressures of 1 and 4 kbar showed that the phenakite solubility in granitic melts increases with the growth of their alkalinity (agpaitic index), temperature, and pressure. It was also established that the BeO content in glass in the presence of water fluid is higher than that in dry conditions. In particular, at 1100°C the water-bearing melts contain ~7 wt % BeO at $K_{\text{agp}} \sim 2.3$, while dry melts contain ~5.4 wt % at $K_{\text{agp}} \sim 2.1$ (Table 3). These values are much higher than the average beryllium content (~1.5 wt % BeO) in the highest grade Ermakovskoe fluorine–beryllium hydrothermal deposit.

According to studies (Wood, 1992; Barton and Young, 2002), Be in hydrothermal conditions migrate mainly with fluorine-bearing complex compounds. However, experimental studies in the water–granite melt system (London et al., 1988) showed that Be has the greater affinity to the melt than to a fluid phase; the Be partition coefficient is $K_D = 0.81$. However, in the given case, experiments were carried out with a pure water instead of water–salt fluid, which is usually typical of the natural systems of beryllium deposits. Direct determinations of Be concentration in the magmatic fluid inclusions are scarce. For instance, they were obtained by AES-LM method at the Geological Institute of the Siberian Branch of the Russian Academy of Sciences (Ishkov and Reyf, 1990; Reyf and Ishkov, 1999; Reyf, 1997, 2004). In the magmatic

fluids of the Ermakovskaya granite intrusion, the measured Be concentrations (C_{Be}) account for 0.04–1.8 g/kg solution, averaging 0.28 g/kg (0.028 wt %). Similar C_{Be} values (0.06 to 2.5 g/kg or 0.006–0.025 wt %) were established in fluids related to the Be-bearing miarolitic pegmatites of the Pamirs (Prokofiev et al., 2003), whereas C_{Be} in the magmatic brine inclusions of the Mariiktikan tungsten-bearing intrusion, based on unpublished data by F.G. Reyf, seemed to be lower than 0.005 g/kg (~0.0005 wt %).

Experimental and calculated data according to (Wood, 1992) showed that Be in hydrothermal fluids is transferred mainly as fluoride and fluorine–carbonate complexes, while its solubility in the fluorine-bearing fluids is several orders of magnitude higher than in systems with other salt components. Experiments performed by (Evensen et al., 1999) also demonstrate that the Be contents in the melt increase in the fluorine-bearing systems. An addition of F in the granitic melt at our experimental parameters rather would provide even greater BeO concentration in melts.

The generalization of our experimental data and data by (Evensen et al., 1999) made it possible to deduce the dependence of the BeO solubility in water-bearing granite melts on temperature, pressure, and alkalinity (agpaitic index) in “saturated” (based on Be in the presence of phenakite crystals) conditions:

$$\log(\text{BeO, wt \%}) = a_1 + b_1 K_{\text{agp}} + c_1 (K_{\text{agp}})^2 + (a_2 + b_2 K_{\text{agp}} + c_2 (K_{\text{agp}})^2) P, \quad (9)$$

where:

$$a_1 = -21.336 + 58.729(1000/T) - 41.519(1000/T)^2;$$

$$b_1 = 27.791 - 72.294(1000/T) + 54.475(1000/T)^2;$$

$$c_1 = -10.170 + 27.266(1000/T) - 18.537(1000/T)^2;$$

$$a_2 = 5.198 - 11.5097(1000/T) + 5.89331(1000/T)^2;$$

$$b_2 = -3.276 + 7.441(1000/T) - 3.734(1000/T)^2;$$

$$c_2 = 0.008568 - 0.09862(1000/T)^2;$$

T is the temperature, K; P is the pressure, kbar; $K_{\text{agp}} = (\text{Na} + \text{K})/\text{Al}$ is the agpaitic index.

The equation was obtained from 49 points (runs), including 23 points from work (Evensen et al., 1999) within a temperature interval of 700–1100°C, a pressure interval of 1–4 kbar, and $K_{\text{agp}} = 0.7$ –2.3. The root mean square deviation is 0.1, coefficient of variation 12%, error (Ex) = 0.03. The region of determination: temperature interval of 600–1100°C, pressure interval of 1–4 kbar, agpaitic index from 0.6 to 1.3 at 600–850°C and from 0.7 to 2.2 at 900–1100°C.

Figure 5 demonstrates a good agreement of experimental results and calculated data according to equation (9). Thus, this equation could be used for estimating the BeO concentration in the water-bearing Be-saturated granitic melts within indicated intervals of temperature, pressure, and compositions of melts.

Obtained results are consistent with data on the higher Be productivity of more water-saturated granitoids (Reyf, 2008). The revealed dependence of the Be solubility on the melt alkalinity indicates that alkaline granites are the most productive rocks for beryllium mineralization. Thereby, the levels of Be concentration in the alkaline aluminosilicate melts could reach values exceeding Be content in the high-grade ores of hydrothermal deposits.

We showed that the Be solubility positive correlates with temperature, but shows less dependence on pressure. Nevertheless, the Be solubility slightly increases at higher pressures. Hence, at cooling and crystallization of the melt, the dissolved Be will be released and redistributed in a fluid phase.

Obtained experimental data indicate the possibility of significant Be accumulation in residual alkaline hydrous melts. Given the fact that the presence of fluorine could lead to the increase of Be solubility, the absolute values of concentrations of this element in melts will be much higher than the beryllium content in the high-grade ores. Hence, comparatively small alkaline–granitoid bodies could produce large Be deposits, formation of which will be controlled by structural, lithological, and other factors. Be-rich hydrous alkaline–granitoid melts likely are formed through decompression-related heterogenization in the fluid–magmatic systems during evolution of granitic melts (Kotelnikov et al., 2019; Shapovalov et al., 2019).

Application of Experimental Data on the Genesis of the Ermakovskoe Deposit

Obtained experimental data that demonstrated the dependence of the Be solubility on the melt alkalinity allow us to regard the alkaline granites as the most productive for the beryllium mineralization. This can be exemplified by the Russia’s largest Ermakovskoe F–Be deposit, which is genetically related to the alkaline granite intrusion (Damdinova et al., 2018; Daminova and Reyf, 2004; Lykhin and Yarmolyuk, 2015; Reyf, 2008; Kupriyanova and Shpanov, 2011, and others). Our data show that the alkaline melt could contain sufficiently high Be concentration, which is higher than its concentration in the ores of the considered deposit, where the average BeO content is ~1.5 wt %. It is known that the beryllium content in natural granitoids could reach 160 ppm only in the most enriched varieties, while clarke values of this element in the granites vary from 3 to 28 ppm (London and Evensen, 2002). Thereby, the experimentally obtained Be content could reach over 7 wt % BeO.

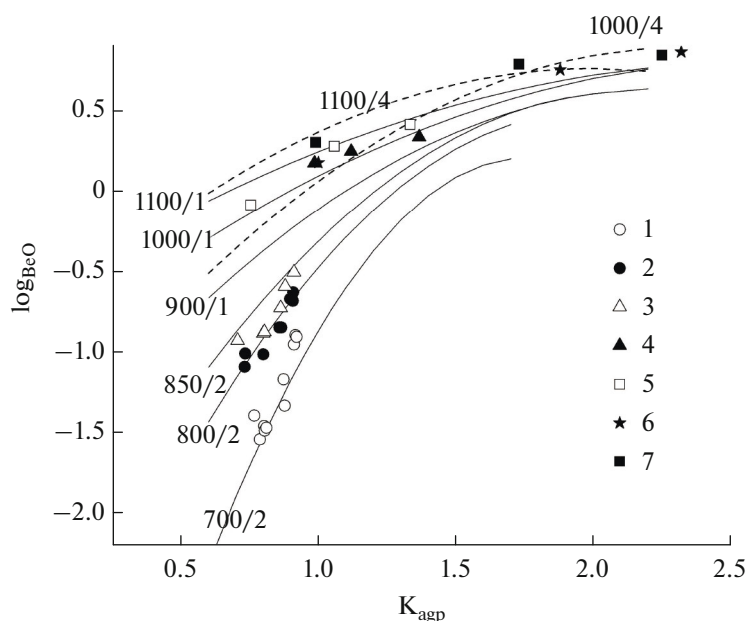
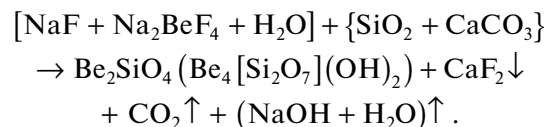


Fig. 5. Comparison of experimental results and calculated data on the phenakite solubility in aluminosilicate melt. Data from (Evensen et al., 1999): (1) at 700°C and 2 kbar, (2) at 800°C and 2 kbar, (3) at 850°C and 2 kbar; our data: (4) at 1000°C and 1 kbar, (5) at 1000°C and 4 kbar, (6) at 1100°C and 1 kbar, (7) at 1100°C and 4 kbar. Solid lines are curves for pressure of 1 and 2 kbar calculated from equation (9); dashed curves are those for a pressure of 4 kbar. Temperature in °C in numerator, pressure in kbar, in denominator.

Since the accumulation of beryllium in natural deposits is always assisted by fluid both in pegmatites forming from a fluid-rich residual melts and at the hydrothermal deposits of obviously fluid-magmatic genesis, beryllium as typical incompatible element could partially escape in fluid phase. As was shown previously (Wood, 1992), beryllium in hydrothermal conditions migrate mainly with fluorine-bearing complexes. Therefore, the presence of fluorine in a fluid is the main condition for the beryllium extraction into the fluid phase. In this case, the fluid–melt partition coefficient also will significantly increase. As was shown by experimental studies in the NaF–H₂O system, the fluorine solutions are solutions of the second (P–Q) type, which are characterized by the presence of two regions of phase immiscibility (Kotelnikova and Kotelnikov, 2008). At *P–T* parameters of the onset of crystallization of granitoid melt, fluoride water–salt fluid is immiscibly split into concentrated alkaline sodium–fluoride fluid and diluted NaF-bearing aqueous fluid. The study of melt and fluid inclusions in the ore-generating granitoids of the Orotskoe and Ermakovskoe beryllium deposits confirmed the existence of such immiscible fluid phases (Reyf and Ishkov, 2003; Reyf, 2008). Thereby, the total concentration of salt components in the highly concentrated fluids represented by the fluorine-bearing sulfate–chloride phases could reach 70–75 wt %. This phase provides the maximum extraction of incompatible elements, including Be, from granitoid melt. Later, Be-bearing sodic solutions with the high fluorine content

penetrate the host carbonate–silicate rocks and form characteristic assemblages of Be minerals, phenakite and bertrandite (Be₄[Si₂O₇](OH)₂), with fluorite:



One of the main factors in the formation of high-grade beryllium ores is the favorable lithological composition of host rocks of ore-bearing intrusion, as well as alkaline-acid conditions of the migration of metal-liferous fluids (Damdinova and Reyf, 2004, 2008).

CONCLUSIONS

(1) It was experimentally established that the solubility of phenakite in the aluminosilicate melts of different composition increases with increasing alkalinity (agpaitic index) of melt, temperature, and pressure. Thereby, the BeO content in the glass in the presence of water fluid is higher than in dry conditions.

(2) It was shown that beryllium is characterized by the high solubility in granitic melts, where BeO content could be sufficiently high to form the high-grade deposits.

(3) Obtained experimental data indicate in support of the Be accumulation in residual water-bearing alkaline melts, which are formed at decompression-related

heterogenization in fluid-magmatic systems during evolution of granite melts.

(4) Experimental data are generalized as equations describing the BeO solubility in water-bearing alkaline granite melts in the Be-“saturated” conditions (in equilibrium with Be crystalline phases) depending on the agpaite index, temperature, and pressure.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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